

Image Reconstruction for Magnetic Particle Imaging Based on Sparse Representation and Deep Learning

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Abstract—The performance of the image reconstruction method is crucial in magnetic particle imaging (MPI) as it helps transform the raw signal data into a quantitative representation of the nanoparticle distribution within the imaging area. In this study, an image reconstruction method based on sparse representation and deep learning (SR-DL) is proposed to solve the inverse problem of MPI and obtain high-quality MPI images. The sparse-representation (SR) method we proposed earlier is used to obtain the initial MPI image. Then the residual encoder-decoder convolutional neural network (RED-CNN) structure is constructed to enhance the initially reconstructed image. A fast dataset generation method is proposed by combining the simulated phantom, measured system matrix, and measured noise data, which can be easily ported to other MPI systems. Numerical simulations are carried out to evaluate the performance of the proposed method based on different loss functions. Furthermore, phantom experiments are conducted using a custom-built narrowband MPI scanner with a gradient of 1.1 T/m in x - and y -directions to assess the feasibility of the SR-DL method. Compared with the SR method, the SR-DL method improves the spatial resolution from 0.5 to 0.3 mm in the x -direction and from 1.2 to 0.8 mm in the y -direction and reduces the artifacts. We envisage that the SR-DL method is significant to improve the image quality and reduce the hardware consumption of MPI.

Index Terms—Deep learning (DL), image reconstruction, magnetic particle imaging (MPI), modulation transfer function (MTF), system function.

I. INTRODUCTION

MAGNETIC particle imaging (MPI) is a promising imaging modality with applications to biomedicine, including vascular imaging [1], tumor detection [2], and drug monitoring [3]. The MPI system is used to obtain the spatial distribution of the magnetic nanoparticles (MNPs) using the magnetization response of the MNPs [4]. Unlike traditional imaging techniques such as magnetic resonance imaging (MRI) or computed tomography (CT), the MPI measurement

is quantitative, as the MPI signal directly represents how much MNPs exist at certain positions. The advantages of MPI also include high spatial and temporal resolution and high sensitivity [5], [6], [7].

The image reconstruction process in MPI plays a pivotal role in transforming acquired raw data into high-resolution images, enabling accurate representation of nanoparticle distribution within the imaged region. The performance of the image reconstruction method has a significant impact on image quality [8], [9], [10], which can be quantitatively evaluated by some evaluation indexes, such as structural similarity (SSIM) index, peak signal-to-noise ratio (PSNR), and spatial resolution. High-quality MPI images can provide an accurate scientific basis for medical decisions and are of significant importance in biomedical applications [11], [12].

The objective of the image reconstruction in MPI is to determine the particle distribution in space using the measured magnetization signal, which is an ill-posed inverse problem. The Tikhonov regularization, alternating direction method of multipliers (ADMMs), and algebraic reconstruction technique (ART) methods are the most widely used approaches for system matrix-based reconstruction [13], [14]. To address issues such as noise and edge preservation, Storath et al. [15] developed an algorithmic framework using fused lasso solvers based on regularization strategies. Ilbey et al. [16] have shown that the ADMM method, which involves total variation and L_1 norm minimization, can significantly enhance image quality. Nonetheless, the non-differentiability of the L_1 norm makes calculations more resource-intensive.

In recent years, image reconstruction methods based on deep learning (DL) techniques for MPI have been reported. In 2017, Chae [17] proposed a method to reconstruct MPI images using a single-layer fully connected network. In 2020, Dittmer et al. [18] proposed an image reconstruction framework using deep image prior (DIP). Shang et al. [19] constructed a CNN model to reconstruct the MPI image. Wu et al. [20] proposed a projection generation network (PGNet) to overcome the sparse-view problem in MPI. Güngör et al. [21] proposed a physics-driven method for MPI reconstruction based on the deep equilibrium model with learned data consistency (DEQ-MPI).

The deep-learning-based reconstruction method can generate high-quality images, increase reconstruction speed, and reduce the noise [22]. However, DL models require a large amount of high-quality training data. Gathering and annotating

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such data can be time-consuming and expensive, and the performance of the model heavily relies on the quality and representativeness of the training dataset [23]. In addition, DL models are often considered “black boxes,” which means that it is challenging to understand and interpret the decision-making process of the model. This lack of interpretability can be a concern in critical applications, such as healthcare.

In this work, we propose a framework for MPI image reconstruction by combining the traditional method and DL method. In this framework, the sparse-representation (SR)-based method is used to obtain the initial MPI image. Then, the RED-CNN network structure is constructed to enhance the quality of the initial image.

The main contribution of this article can be summarized as follows:

- 1) An image reconstruction method based on sparse representation and DL (SR-DL) is proposed to improve the quality of MPI images. Compared with our previous work in [24], the resolution in the unit of mT is further improved with a much lower gradient magnetic field, which is beneficial to reduce the power consumption and increase the field of view (FOV).
- 2) A fast dataset generation method is proposed by combining the simulated phantom, measured system matrix, and noise data to train the neural network. This generation method can be easily ported to other MPI systems, which improves the universality of the proposed method.

II. METHODOLOGY

A. Basic Principle of MPI

The basic principle of MPI is based on the response of MNPs to an external magnetic field, which is called the excitation field and is usually sinusoidal

$$H^E(t) = A^E \cos(2\pi f^E t) \quad (1)$$

where A^E and f^E are the amplitude and frequency of the excitation field, and t represents the time. To obtain the particle concentration in a small voxel in Ω , a highly inhomogeneous selection field is generated, which contains a special location named the field-free point (FFP). The strength of the magnetic field is zero at the FFP position. When applying the selection field, the particle magnetization will be saturated in most regions except a small region around FFP.

By superimposing the excitation field on the selection field, only the MNPs in the near area around FFP react on the excitation field. Then the spatial encoding can be achieved by moving FFP over the FOV using a scanning field.

Taking the 2-D MPI as an example, to detect the change of the particle magnetization, the receive coils are used and the induced voltage $u(t)$ is

$$u(t) = \int_{\Omega} a(x, y, t) c(x, y) dx dy \quad (2)$$

where $(x, y) \in \Omega$ is the spatial position, $c(x, y)$ is the particle concentration, and $a(x, y, t)$ is the system function in time-space, which can be expressed as

$$a(x, y, t) = -\mu_0 s(x, y) \frac{\partial}{\partial t} m(x, y, t) \quad (3)$$

where μ_0 is the vacuum permeability, $s(x, y)$ is the sensitivity of the receive coil, and $m(x, y, t)$ is the mean magnetization response of particles.

Then by analyzing the spectrum, the measured voltages u_k at each frequency component k are [16]

$$u_k = \int_{\Omega} a_k(x, y) c(x, y) dx dy, \quad k = 1, \dots, K \quad (4)$$

where $a_k(x, y)$ is the system function in frequency space, k represents the harmonic order, K is the total number of frequencies, and $c(x, y)$ represents the particle concentration.

For the single-harmonic-based narrowband MPI approach, only the single harmonic of the measured signal is used for imaging, e.g., the third harmonic u_3 . u_3 measured at the FFP position can be written as a convolution of $h(x, y)$ and the point spread function (PSF) [25]

$$u_3(x_{\text{FFP}}, y_{\text{FFP}}) = h(x, y) * \text{PSF}(x, y) \big|_{(x,y)=(x_{\text{FFP}}, y_{\text{FFP}})} \quad (5)$$

where $h(x, y)$ is the third harmonic generated by the local MNPs at position $(x_{\text{FFP}}, y_{\text{FFP}})$, which is proportional to the concentration $c(x, y)$.

After discretization, the relationship between the measured signal and particle concentration can be described by a linear integral equation and written in a matrix form

$$U = AC \quad (6)$$

where $U \in \mathbb{C}^{(L,1)}$ is composed of the measured voltages, $A \in \mathbb{C}^{(L,M \cdot N)}$ is the system matrix, and $C \in \mathbb{C}^{(M \cdot N,1)}$ is the particle concentration distribution. L is the number of measurements. M denotes the number of image pixels in the x -direction and N denotes the number of image pixels in the y -direction.

B. MPI Image Reconstruction

The MPI image reconstruction process is to obtain the particle concentration from the measured voltages by solving (5). In this work, an image reconstruction method based on SR-DL is proposed. The framework of the proposed method is shown in Fig. 1. The initial MPI image is reconstructed using the SR method. Then, the initial image is input to the DL network to further improve the image quality.

1) *Initial Image Reconstruction*: The SR method is applied to obtain the initial MPI image. First, the concentration distribution is decomposed into several basis concentration elements, which can be expressed as

$$C = GW \quad (7)$$

where $G = \{g_1, g_2, \dots, g_L\}$ is composed of the basis elements, $W = [\omega_1, \omega_2, \dots, \omega_L]^T$ represents the weight coefficient vector and L represents the number of the basis elements. Then, (6) is rewritten as

$$U = AGW. \quad (8)$$

The Gaussian radial basis function is used as the basis element and the Gauss-Newton method is used to solve (8)

$$W_{q+1} = W_q - \alpha(B^T B + \beta I)^{-1}(B^T(B \cdot W_q - U) + \beta I \cdot W_q) \quad (9)$$

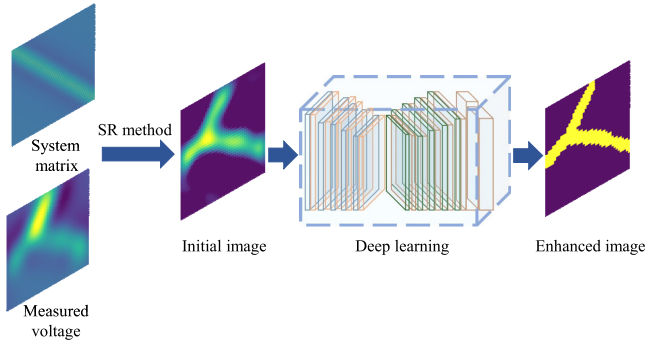


Fig. 1. Main framework of the proposed image reconstruction method.

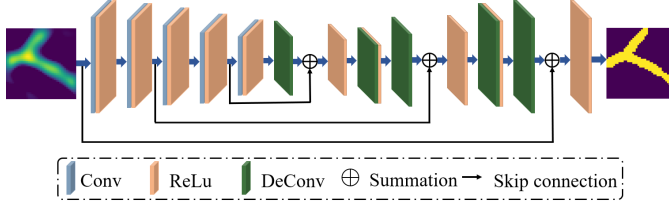


Fig. 2. Main framework of the RED-CNN network model.

where $B = AG$, q , α , β , and I represent the iteration number, iteration step, the regularization parameter, and the identity matrix, respectively. More details of the SR method are shown in [24].

2) *Image Enhancement Using DL*: The RED-CNN network model is used to enhance the image quality, which consists of five convolutional layers and five deconvolution layers, with each layer followed by a corrected linear unit (ReLU), as shown in Fig. 2.

The calculation formula for convolution processing is

$$H_{\text{out}} = F((H_{\text{in}} - K_s + 2 \cdot p_d)/s + 1) \quad (10)$$

where H_{out} is the width or length of the output image, F is the convolution function for DL, H_{in} is the width or length of the input image, p_d is the padding amount of the convolution, K_s is the size of the convolution kernel, and s is the convolution stride. After performing a convolution operation, the feature map is obtained while the length and width of the feature map generally decrease. Deconvolution is equivalent to the inverse process of convolution, and its processing formula is

$$H_{\text{out}} = F'(H_{\text{in}} - 1) \cdot s - 2 \cdot p_d + K_s \quad (11)$$

where F' is the deconvolution function of DL. After performing the five deconvolution operations, the length and width of the image gradually increase until it is the same as the input size. In addition, three skip connections are added to the network to avoid gradient vanishing or exploding.

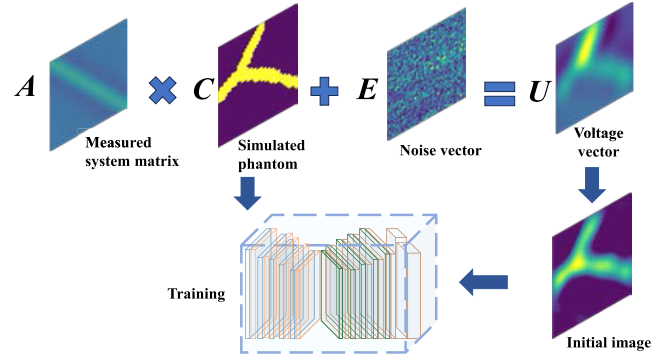


Fig. 3. Framework of the proposed fast dataset generation method.

III. TRAINING OF THE DL MODEL

A. Generation of the Training Dataset

Due to the large amount of data required for training neural networks, obtaining a sufficient amount of actual MPI measurement data is difficult. In this work, a fast dataset generation method is proposed by combining the simulated phantom, measured system matrix, and noise data, as shown in Fig. 3. The training set can be obtained in the following three steps:

1) *Measurement of System Matrix and Noisy Data*: To obtain the system matrix, the solution of MNPs (Synomag¹ D-70) is injected into a point phantom model. The iron concentration of the solution is 10 mg/mL. Then the signal acquisition is performed using the single-harmonic MPI scanner. In x - and y -directions, the gradient intensity is 1.1 T/m, and that in z -direction is 2.2 T/m. The strength and frequency of the excitation field are 4 mT and 10 kHz, respectively. In the experiment, Cartesian trajectories are used for FFP scanning.

The FFP is changed in the imaging area using the scanning field generated by the scanning coils, as shown in Fig. 4(a). At each FFP position of $(x_{\text{FFP}}, y_{\text{FFP}})$, the magnetization response signal $u(x_{\text{FFP}}, y_{\text{FFP}})$ is collected when the position of the sample (x, y) is $(0, 0)$. According to the principle of the single-harmonic MPI approach in [25] and described above, the third harmonic of the collected response signal $u_3(x_{\text{FFP}}, y_{\text{FFP}})$ is extracted for image reconstruction, as shown in Fig. 4(b).

Then the 2-D PSF can be obtained, as shown in Fig. 5(a). The PSF matrix is transformed to obtain the system matrix, which is expressed as in, (12), shown at the bottom of the page, where $\text{PSF}(x, y)$ is the value at the position (x, y) of the PSF matrix, $\Delta x = (x_M - x_1)/(M - 1)$ and $\Delta y = (y_N - y_1)/(N - 1)$.

¹Registered trademark.

$$A = \begin{bmatrix} \text{PSF}(0, 0) & \text{PSF}(\Delta x, 0) & \cdots & \text{PSF}((M - 1)\Delta x, (N - 1)\Delta y) \\ \text{PSF}(-\Delta x, 0) & \text{PSF}(0, 0) & \cdots & \text{PSF}((M - 2)\Delta x, (N - 1)\Delta y) \\ \vdots & \vdots & \vdots & \vdots \\ \text{PSF}(-(M - 1)\Delta x, -(N - 1)\Delta y) & \text{PSF}(-(M - 2)\Delta x, -(N - 1)\Delta y) & \cdots & \text{PSF}(0, 0) \end{bmatrix} \quad (12)$$

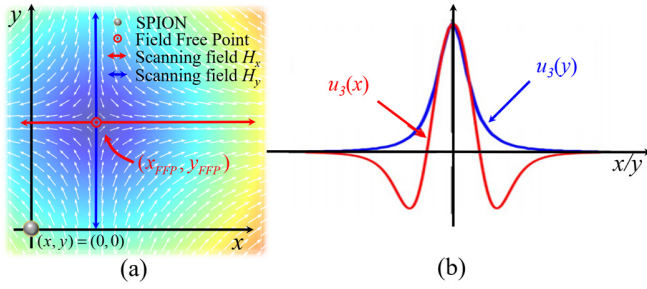


Fig. 4. (a) Illustration of the movement of FFP position and (b) third harmonic of the collected response signal. (Note: In the built MPI scanner, the FFP movement in x-direction is implemented by moving the MNP sample mechanically. The results obtained using the electrical and mechanical scanning are equivalent).

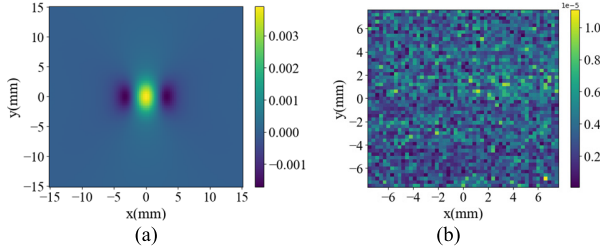


Fig. 5. Measured PSF and noisy data using the MPI scanner. (a) Measured PSF. (b) Measured noise data.

Then the MNP sample is removed, and the third harmonic of the signal is extracted to form the noise vector E , as shown in Fig. 5(b).

2) *Calculation of Voltage Vector*: The vascular mimics based on digital retinal images (DRIVE) are extracted to simulate the particle concentrations. The DRIVE dataset contains 40 manually segmented retinal vascular images with a pixel size of 768×584 . These images are divided and converted to 11 763 MPI images with a size of 51×51 . To improve the enhanced performance of small targets, 2879 MPI images of point phantoms with random positions and diameters are added.

Then the voltage vector for different phantoms can be calculated by

$$U = AC + E \quad (13)$$

where A is the system matrix measured in the experiment, C is the simulated phantom distribution, and E is the measured noise data.

For each image, the initial concentration is sequentially diluted at 1.78^p times, which is equivalent to $p \cdot 5$ dB, and $p = 0, 1, 2, \dots, 5$. Then the SNR of the voltage vector will decrease by 25 dB from the highest SNR at 5 dB intervals. In this way, the number of MPI images and corresponding voltage vectors is expanded to 87 852.

3) *Reconstruction of Initial MPI Images*: Using the SR method, a total number of 87 852 initial MPI images are preliminarily reconstructed from the calculated voltage vector and the measured system matrix. Finally, the datasets containing 87 852 samples are created for DL network training. Each sample includes an initial MPI image and a reference image.

B. Training of the RED-CNN Network

About 80% of the created dataset is used as the training dataset, and the remaining 20% is used as the testing set to test the performance of the RED-CNN network. When training the neural network model, the initial MPI images are used as the input of the neural network, and the reference images are used as the label. The convolution kernel size is set to 5×5 . The number of filters in the last layer is 1, the number of filters in other layers is 96, the stride of convolution and deconvolution is set to 1 with no padding, and the learning rate is 2×10^{-4} . The number of epochs used to train the RED-CNN model is set to 200, and the mini-batch size is 2000.

Two different types of loss functions are selected for training the RED-CNN network, which are mean square error (MSE) and mean absolute error (MAE). In this study, the loss functions are both optimized by the Adam. MSE can be expressed as

$$\text{MSE} = \frac{1}{M \cdot N} \sum_{m=1}^M \sum_{n=1}^N \|C_r(x_m, y_n) - \Phi(C_i(x_m, y_n))\|_2^2 \quad (14)$$

where $C_i(x_m, y_n)$ and $C_r(x_m, y_n)$ represent the pixel values of the initial image and the reference image at position (x_m, y_n) , respectively. Φ represents the mapping function from the initial reconstructed MPI images to the images enhanced by the trained model. The MSE is the sum of the squares of the distance between the predicted value and the true value. When using the MSE as the loss function, the convergence rate is fast but the edges of the processed image may become blurred.

MAE can be expressed as

$$\text{MAE} = \frac{1}{M \cdot N} \sum_{m=1}^M \sum_{n=1}^N |C_r(x_m, y_n) - \Phi(C_i(x_m, y_n))| \quad (15)$$

which is the sum of the absolute values of the difference between the predicted value and the true value. When using the MAE as the loss function, the convergence rate is slow but the edges of the processed image are clearer.

IV. SIMULATION RESULTS

In the simulation, the proposed SR-DL method based on different loss functions was used to reconstruct the image of the point phantoms with different distances d in the x- and y-directions, as shown in Fig. 5. The diameter of the point phantom was 1.5 mm. The distance d decreased from 1.5 to 0.3 mm at an interval of 0.3 mm. For each phantom, the corresponding simulated voltage vector was calculated using (13). The SR method was used to calculate the initial MPI image. Then the initial MPI images were input into the trained network model based on MSE and MAE, respectively. The output of the neural network was the final enhanced MPI image.

Fig. 6 shows that the initial MPI image produces blurring at the edges of the object. After being enhanced by the SR-DL method, the size and shape of the phantoms were more accurate and closer to the reference image, and the edges were clearer.

When the loss function was selected as MAE, the uniformity of the output image of the proposed method was better. When

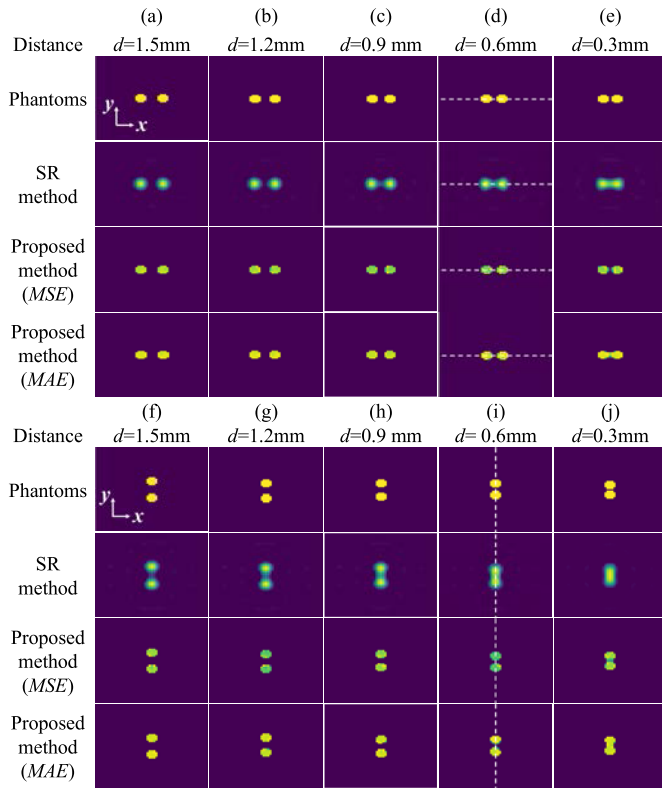


Fig. 6. MPI images reconstructed using different methods with different distances d in (a)–(e) x -direction and (f)–(j) y -direction.

the distance d in the x - and y -directions was 0.6 mm, the edges of the two objects in the images reconstructed by the SR method became adhesive and could not be clearly distinguished. When the distance was reduced to 0.3 mm, the two objects could still be distinguished using MSE, but failed to be distinguished using MAE.

For the point phantom with $d = 0.6$ mm, Fig. 7 showed the extracted pixel values on the white lines in Fig. 6. After enhanced by the proposed method, the 1-D curve of the image became sharper at the edges, resulting in higher contrast. In the distribution area of MNPs, the uniformity became higher, and the concentration values were consistent with the reference values. In addition, the proposed method reduced the artifacts of the initial MPI image.

For a quantitative evaluation, the SSIM and PSNR of the reconstructed MPI images were calculated, which were displayed in Fig. 8. The images reconstructed using the SR-DL method had higher SSIM and PSNR compared to the initial images reconstructed using the SR method, which verified the effectiveness of the proposed framework. In addition, for most phantoms, the SSIM and PSNR of the images reconstructed using the proposed method based on MAE were higher than those based on MSE.

To evaluate the spatial resolution of the reconstructed images, the modulation transfer function (MTF) was calculated. Variation of the MTFs of the reconstructed images with the spatial frequencies in the x - and y -directions were shown in Fig. 9. As the spatial frequency increased, the MTF gradually decreased. The spatial resolution was defined as

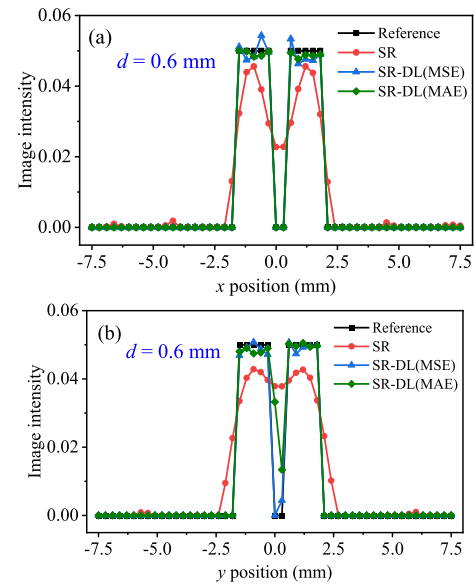


Fig. 7. Pixel values on the white lines using the SR and the proposed methods in (a) x -direction and (b) y -direction when $d = 0.6$ mm.

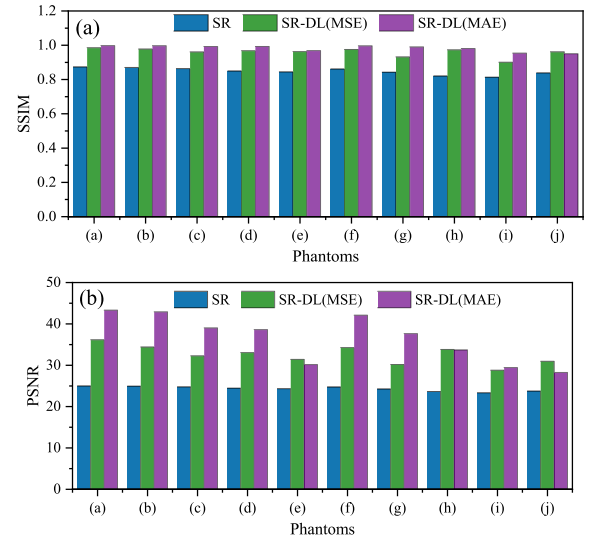


Fig. 8. Comparison of the (a) SSIM and (b) PSNR using the SR and the proposed methods.

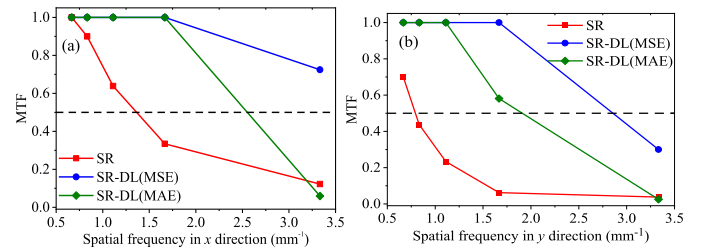


Fig. 9. Variation of the MTFs of the reconstructed images with the spatial frequencies using the SR and the proposed methods in (a) x -direction and (b) y -direction.

the reciprocal of the corresponding spatial frequency when the MTF was 0.5. It showed that the images reconstructed using the proposed method based on MSE had higher spatial resolution than those based on the MAE and the SR method.

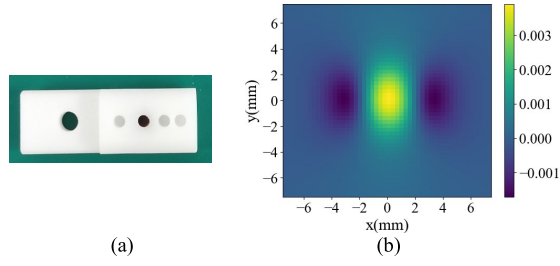


Fig. 10. Point phantom and the measured original signal.

V. EXPERIMENTS AND DISCUSSION

A. Experiments Description

The experiments were performed using the developed MPI scanner described above. The third harmonic of the magnetic nanoparticle signal was measured for image reconstruction, which could be expressed as

$$U = [u_3(x_{\text{FFP}} = x_1, y_{\text{FFP}} = y_1) \cdots u_3(x_{\text{FFP}} = x_M, y_{\text{FFP}} = y_N)]^T \quad (16)$$

where $u_3(x_{\text{FFP}} = x_m, y_{\text{FFP}} = y_n)$ was the third harmonic of the measured signal when the position of FFP was (x_m, y_n) . The pixel size of the reconstructed MPI image was 0.3×0.3 mm. The imaging area was 15×15 mm. Therefore, the number of pixels in the reconstructed image was 51×51 .

Fig. 10(a) shows the constructed point phantom. After performing the MPI acquisition, the measured original signal U using the narrowband MPI scanner is shown in Fig. 10(b). The SNR could be calculated according to the measured signal U and the measured noise data E , which could be expressed as

$$\text{SNR} = 20 \log \frac{U}{E}. \quad (17)$$

When the iron concentration of the solution was 10 mg/mL, the calculated maximum SNR was 85.8 dB.

B. Reconstructed Results of Point Phantoms

The point phantoms with different distances were constructed in the experiments, as shown in Figs. 11 and 12. The distance d ranged between 0.3 and 1.5 mm in x -direction and between 0.6 and 2 mm in y -direction. The diameter of the point phantom was 1.5 mm, and the depth was 3 mm.

The MPI images reconstructed using the SR and the proposed methods for different phantoms in x -direction were shown in Fig. 11, and those in y -direction were shown in Fig. 12. It was shown that the images reconstructed using the SR method were relatively blurry at the edge of the phantom, while after enhanced by the proposed method, the reconstructed point phantoms were clearer at the edge of the phantom, and could provide more accurate shape and size.

In terms of artifacts, a small amount of artifacts existed in the images reconstructed by the SR method. After being enhanced by the proposed method based on MAE, almost no artifacts existed in the background area.

The pixel values on the white lines in Figs. 11 and 12 were shown in Fig. 13. After enhanced by the proposed method,

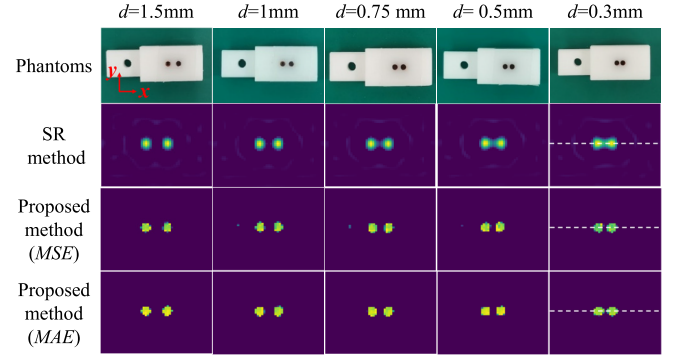


Fig. 11. MPI images reconstructed using different methods with different distances in x -direction.

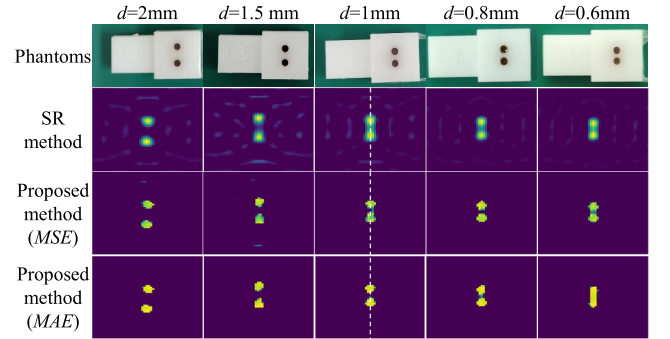


Fig. 12. MPI images reconstructed using different methods with different distances in y -direction.

the 1-D curve of the image became sharper at the edges and had better uniformity in the point regions, thus having higher spatial resolution ability for objects. When the distance d in the x -direction was 0.3 mm and the distance d in the y -direction was 1 mm, the two points reconstructed by the SR method were slightly overlapped. However, the output of the proposed method had a higher intensity at the edge of the phantom than that at the center and could still distinguish the two-point objects.

To evaluate the spatial resolution, the MTFs of different reconstructed images with different spatial frequencies were compared. The MTFs calculated using the SR method and the proposed method in the x -direction were shown in Fig. 14(a), and those in the y -direction were shown in Fig. 14(b).

It was indicated that the MTFs calculated using the SR-DL method were higher than those using the SR method. $\text{MTF} = 0.5$ was displayed in Fig. 14 by the black dashed lines. Compared with the proposed method based on MAE and the SR method, when $\text{MTF} = 0.5$, the reciprocal of the corresponding spatial frequency of the SR-DL method based on MSE was higher, illustrating a better spatial resolution. When the gradient was 1.1 T/m in x - and y -directions, the SR-DL method could improve the spatial resolution of the initial image from 0.5 to 0.3 mm in x -direction and from 1.2 to 0.8 mm in y -direction.

In consideration of the gradient strength, the resolution ΔH in the unit of mT was defined as: $\Delta H = G \times \Delta x$. This represented a spatial resolution Δx in the unit of mm at a gradient G .

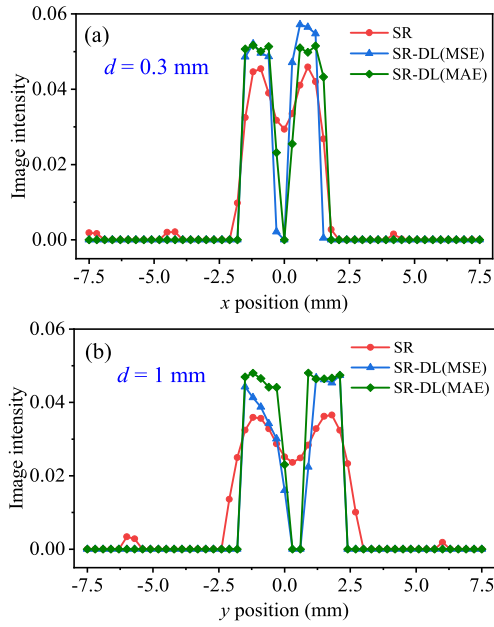


Fig. 13. Pixel values on the white lines using the SR and the proposed methods in the (a) x -direction when $d = 0.3$ mm and (b) y -direction when $d = 1$ mm.

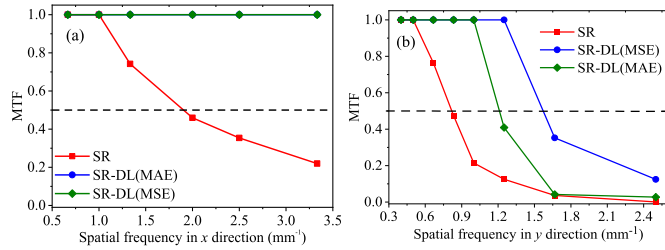


Fig. 14. Variation of the MTFs of the reconstructed images with the spatial frequencies using the SR and the proposed methods in (a) x -direction and (b) y -direction.

TABLE I

SPATIAL RESOLUTION Δx AND RESOLUTION ΔH ACHIEVED USING THE SR AND THE PROPOSED METHODS IN x - AND y -DIRECTIONS USING DIFFERENT MPI SCANNERS

MPI scanner	x direction			y direction		
	TUBS ($G_x=2.2$ T/m)	BUAA ($G_x=1.1$ T/m)		TUBS ($G_y=4.4$ T/m)	BUAA ($G_y=1.1$ T/m)	
Method	SR	SR	SR-DL	SR	SR	SR-DL
Δx	0.3 mm	0.5 mm	0.3 mm	0.3 mm	1.2 mm	0.8 mm
ΔH	0.66 mT	0.55 mT	0.33 mT	1.32 mT	1.32 mT	0.88 mT

To make a comparison with our previous work [24], the spatial resolution Δx and resolution ΔH achieved using the SR and the proposed method using different MPI scanners are shown in Table I.

In our previous work, the gradient was 2.2 T/m in the x -direction and 4.4 T/m in the y -direction, Δx obtained using the SR method was 0.3 mm in both directions. The resolution ΔH achieved using the SR method in the x -direction was $\Delta H = G_x \times \Delta x = 2.2$ T/m $\times$ 0.3 mm = 0.66 mT, and that in the y -direction was 1.32 mT.

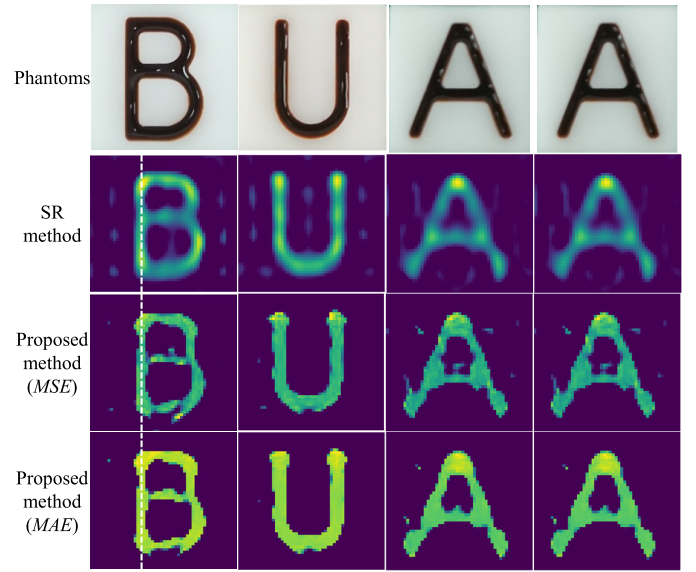


Fig. 15. MPI images obtained using the SR and proposed methods for phantoms of the "BUAA" logo.

In this article, the gradient became 1.1 T/m in x - and y -directions, the spatial resolution achieved using the SR method was 0.5 mm in x -direction and 1.2 mm in y -direction, while that using the proposed method was 0.3 mm in x -direction and 0.8 mm in y -direction. Therefore, the resolution ΔH achieved using the SR method was 0.55 mT in the x -direction and 1.32 mT in the y -direction, while that using the SR-DL method was 0.33 mT in the x -direction and 0.88 mT in the y -direction.

The calculated resolution ΔH showed that, compared with the SR method [24], the proposed method decreased the resolution ΔH from 0.66 to 0.33 mT in the x -direction and 1.32 to 0.88 mT in the y -direction.

On the other hand, the spatial resolution Δx using the SR-DL method and our previous work in x -direction was both 0.3 mm. However, the gradient in this study was much lower, this helped reduce the power consumption for the scanning field and was beneficial for MPI with a larger FOV.

C. Reconstructed Results of "BUAA" Logo

More complex phantoms of the four letters "BUAA" were used to further evaluate the performance of the SR-DL method. The reconstructed images are shown in Fig. 15. It was shown that the images reconstructed by the proposed method based on MAE were clearer than those based on the MSE and the SR method, with higher uniformity and fewer artifacts in the surrounding area, resulting in better visual effects.

The pixel values extracted on the white line of the "B" images in Fig. 15 were displayed in Fig. 16. Pixel values obtained using the SR method showed intense change, while the values reconstructed using the proposed method had better uniformity and the curve was sharper at the edges. From -5 to 5 mm in y -direction, the standard deviation of the SR method was 7.36×10^{-3} , while that of the proposed method based on MSE was 4.99×10^{-3} , and that based on MAE was 1.66×10^{-3} .

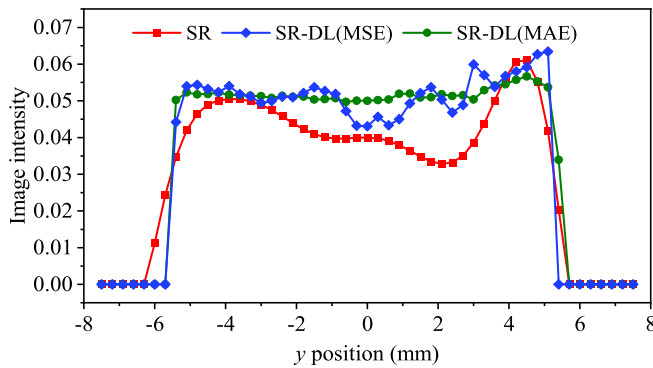


Fig. 16. Pixel values on the white lines of the “B” images.

From Figs. 15 and 16, the proposed method could help obtain MPI images with higher uniformity and resolution. With different loss functions, the characteristics of the MPI images were somewhat different.

MSE penalized larger errors more heavily due to the squared term. In situations where minimizing large errors was crucial, e.g., when $d = 0.6$ mm in Fig. 12, MSE might guide the model to prioritize fitting patterns and features that contributed to reducing these large errors. This might be the reason that the proposed method based on MSE provided higher resolution. However, MSE also tended to amplify errors by squared differences, particularly for outliers.

MAE represented the average of the absolute differences. Compared to MSE, MAE did not amplify errors and treated all errors equally, regardless of their size. Therefore, the MAE could help obtain MPI images with higher uniformity.

VI. CONCLUSION

In this article, an image reconstruction method based on SR-DL was proposed to improve the quality of MPI images. The simulated phantom, measured system matrix, and noise data were combined to provide a fast dataset generation method. This generation method could be easily ported to other MPI systems, which improved the universality of the proposed method.

Phantom experiments with a custom-built narrowband MPI scanner were performed. When the gradient was 1.1 T/m in x - and y -directions, experimental results showed that compared with the SR method, the SR-DL method improves the spatial resolution from 0.5 to 0.3 mm in x -direction and from 1.2 to 0.8 mm in y -direction, and obtain higher SSIM and PSNR and fewer artifacts. In addition, the trained models based on different loss functions (MSE and MAE) were compared. The proposed method based on MAE has higher SSIM and PSNR, while that based on MSE has higher spatial resolution. We believe that the proposed method is anticipated to be highly significant and intriguing for MPI and its biological applications.

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